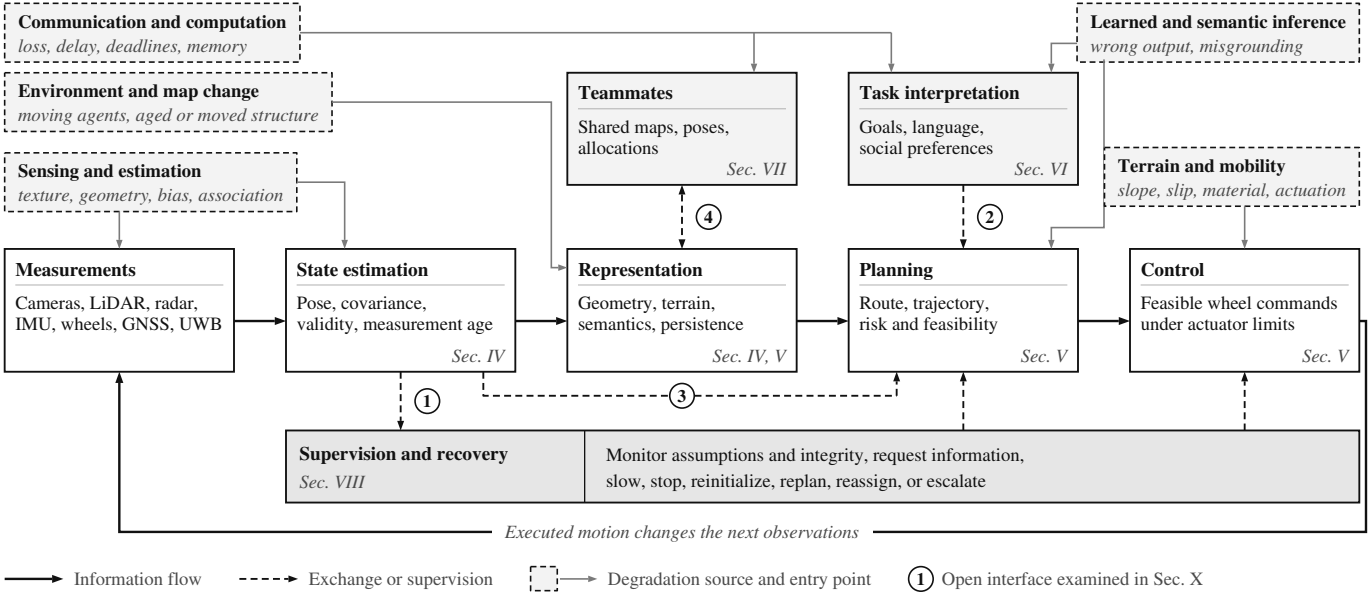

Communication and computation
loss, delay, deadlines, memory
Learned and semantic inference
wrong output, misgrounding
Environment and map change
moving agents, aged or moved structure
Teammates
Shared maps, poses, allocations
Sec. VII
Task interpretation
Goals, language, social preferences
Sec. VI
Terrain and mobility
slope, slip, material, actuation
Sensing and estimation
texture, geometry, bias, association
4
2
Measurements
Cameras, LiDAR, radar, IMU, wheels, GNSS, UWB
State estimation
Pose, covariance, validity, measurement age
Sec. IV
Representation
Geometry, terrain, semantics, persistence
Sec. IV, V
Planning
Route, trajectory, risk and feasibility
Sec. V
Control
Feasible wheel commands under actuator limits
Sec. V
1
3
Supervision and recovery
Sec. VIII
Monitor assumptions and integrity, request information, slow, stop, reinitialize, replan, reassign, or escalate
Executed motion changes the next observations
Information flow
Exchange or supervision
Degradation source and entry point
1 Open interface examined in Sec. X